

The number of two-sided classes of the 0-Hecke monoid is not $p(n)$: a refutation of conjecture 00000003843

Submission to The Last Math Competition

Abstract

Conjecture 00000003843 asserts that the number of two-sided classes of the 0-Hecke monoid $H_n(0)$ equals the partition number $p(n)$, and that the number of left classes inside each two-sided class equals the number of distinct hook lengths of the corresponding partition. We show that $H_n(0)$ is $\mathcal{J}$ -trivial, so it has exactly $n!$ two-sided classes and $n!$ left classes. At $n = 3$ this gives 6 two-sided classes against $p(3) = 3$, and 6 left classes against $\sum_{\lambda \vdash 3} \#\{\text{distinct hook lengths of } \lambda\} = 8$. Both halves of the conjecture therefore fail. The second half already fails at $n = 2$ and is false under the alternative (Kazhdan–Lusztig) reading of the statement as well. Everything is a finite computation; it is formalised in Lean 4 using only the kernel and core, and every computational theorem is verified to depend on no axioms.

1 The conjecture

The conjecture reads, verbatim:

Definition: The 0-Hecke monoid $H_n(0)$ is generated by monoid braid relations and absorption relations. **Conjecture:** The number of two-sided classes of $H_n(0)$ is exactly the partition number $p(n)$, and the number of left classes in each two-sided class equals the number of distinct hook lengths of the corresponding partition. (*0-Hecke two-sided class-partition*)

Throughout, $H_n(0)$ is the monoid generated by $\pi_1, \dots, \pi_{n-1}$ subject to the braid relations, far-commutation, and the absorption relations $\pi_i^2 = \pi_i$. Its elements are indexed by permutations $w \in S_n$, so $|H_n(0)| = n!$, and multiplication by a generator is given by the right-descent rule

$$\pi_w \cdot \pi_i = \begin{cases} \pi_{ws_i}, & \ell(ws_i) > \ell(w), \\ \pi_w, & \text{otherwise,} \end{cases}$$

where ws_i is w with the entries in positions i and $i + 1$ swapped. “Two-sided classes” and “left classes” are read as Green’s $\mathcal{J}$ - and $\mathcal{L}$ -classes: $a \mathcal{J} b$ iff $HaH = HbH$, and $a \mathcal{L} b$ iff $Ha = Hb$. For a finite monoid $\mathcal{D} = \mathcal{J}$, so this is also the two-sided/left cell reading standard in semigroup theory. Section 3 treats the alternative Kazhdan–Lusztig reading.

2 The computation

Building $H_n(0)$ for $n \leq 5$ and computing the principal ideals by closing singletons under left and right multiplication by the generators gives Table 1. Three self-checks are included: the multiplication is verified to be associative on all triples (an exhaustive check, not a sample), the identity

Table 1: Green's classes of $H_n(0)$ against the conjectured values.

n	$ H_n(0) $	$\mathcal{J}$ -classes	$\mathcal{L}$ -classes	$p(n)$	$\sum_{\lambda \vdash n} \#\{\text{distinct hook lengths}\}$
2	2	2	2	2	4
3	6	6	6	3	8
4	24	24	24	5	17
5	120	120	120	7	29

element acts as a two-sided identity, and the number of idempotents comes out as $2, 4, 8, 16 = 2^{n-1}$, which is the known count for $H_n(0)$.

The number of $\mathcal{J}$ -classes equals the number of elements, i.e. every principal two-sided ideal is distinct and $H_n(0)$ is $\mathcal{J}$ -trivial (and hence also $\mathcal{L}$ - and $\mathcal{R}$ -trivial). This is consistent with the literature: Ganyushkin and Mazorchuk prove that H_Γ is $\mathcal{J}$ -trivial [1].

For $n = 3$ the six principal two-sided ideals are, as characteristic vectors on $(\text{id}, s_1, s_2, s_1s_2, s_2s_1, w_0)$:

$$\begin{aligned} J(\text{id}) &= (1, 1, 1, 1, 1, 1), & J(s_1) &= (0, 1, 0, 1, 1, 1), \\ J(s_2) &= (0, 0, 1, 1, 1, 1), & J(s_1s_2) &= (0, 0, 0, 1, 0, 1), \\ J(s_2s_1) &= (0, 0, 0, 0, 1, 1), & J(w_0) &= (0, 0, 0, 0, 0, 1). \end{aligned}$$

They are pairwise distinct, so there are 6 two-sided classes. Note the clean structural reason: $\mathcal{J}$ -triviality means each $\mathcal{J}$ -class is a singleton, so each two-sided class contains exactly *one* left class.

Theorem 1. *The first half of conjecture 00000003843 is false: $H_3(0)$ has 6 two-sided classes but $p(3) = 3$.*

Theorem 2. *The second half of conjecture 00000003843 is false: $H_3(0)$ has 6 left classes in total, whereas the conjecture predicts $\sum_{\lambda \vdash 3} \#\{\text{distinct hook lengths of } \lambda\} = 3 + 2 + 3 = 8$.*

Remark 1. *The two counts in Theorem 1 happen to agree at $n = 2$ ($2 = p(2)$), so $n = 3$ is the smallest witness for the first half. The second half already fails at $n = 2$: there each $\mathcal{J}$ -class is a singleton and so contains one left class, while both partitions of 2 have hook lengths $\{2, 1\}$ and hence 2 distinct hook lengths. Thus $1 \neq 2$ at $n = 2$.*

3 The alternative (Kazhdan–Lusztig) reading

The phrase “the corresponding partition” suggests that the generator had the Kazhdan–Lusztig cell decomposition in mind: for the Hecke algebra of S_n the two-sided cells *are* indexed by partitions, so their number is $p(n)$, and the left cells inside the two-sided cell C_λ are indexed by the standard Young tableaux of shape λ , so there are f^λ of them.

Under that reading the first half of the conjecture is true (for S_n , not for the 0-Hecke monoid) but the second half is still false, because f^λ is the number of standard Young tableaux and not the number of distinct hook lengths. At $n = 2$ we have $f^{(2)} = f^{(1,1)} = 1$, whereas both partitions have hook lengths $\{2, 1\}$, i.e. 2 distinct hook lengths. Already at $n = 2$ the predicted number is 2 while the true number is 1.

So the second half of the conjecture fails under *both* readings, with the same small witness, and the first half fails under the reading forced by the subject of the statement (a monoid, hence Green's relations).

4 Formalisation

The directory `lean4/` contains a Lean 4 project using only the kernel and core (no Mathlib, no sorry). The file `Main.lean` defines

- S_3 in one-line notation together with the inversion-number length, the generators s_1, s_2 , and the right-descent product of $H_3(0)$;
- the principal two-sided and left ideals as closures under multiplication by generators, encoded as characteristic vectors;
- integer partitions, hook lengths, and the count of distinct hook lengths.

The theorems proved are

<code>hSize_eq</code>	$ H_3(0) = 6$,
<code>assoc_ok</code>	associativity on all 6^3 triples,
<code>identity_ok</code>	two-sided identity,
<code>absorption_ok</code>	$\pi_i^2 = \pi_i$ for both generators,
<code>jClasses_eq</code>	$\#\mathcal{J}\text{-classes} = 6$,
<code>lClasses_eq</code>	$\#\mathcal{L}\text{-classes} = 6$,
<code>pCount3_eq</code>	$p(3) = 3$,
<code>sumHooks3_eq</code>	$\sum_{\lambda \vdash 3} \#\{\text{distinct hook lengths}\} = 8$,
<code>refutation_one_bool</code>	$(\#\mathcal{J}\text{-classes} \stackrel{?}{=} p(3)) = \text{false}$,
<code>refutation_two_bool</code>	$(\#\mathcal{L}\text{-classes} \stackrel{?}{=} \text{hook sum}) = \text{false}$,
<code>refutation_part_one</code>	Theorem 1,
<code>refutation_part_two</code>	Theorem 2.

Running `lake env lean Check.lean` prints every computed value and `#print axioms` for every theorem. **All eight computational theorems and both Boolean witnesses depend on no axioms at all.** The two propositional statements `refutation_part_one` and `refutation_part_two` (and the conjunction `summary`) depend on `[propext]` only, which enters through Lean’s generated no-confusion machinery for $\neg$; no `Quot.sound`, `Classical.choice`, `sorryAx` or Mathlib axiom is used anywhere.

One implementation note is worth recording, since it is what makes the axiom-free result possible: Lean 4’s short-circuiting `&&` and `||` elaborate to `if` on a `Bool`, which goes through `CoeSort Bool Prop` and `Decidable (b = true)` and hence pulls in `propext`; and the standard-library list accessor with a default value does the same. Both are therefore replaced here by functions defined directly by pattern matching (`band`, `bor`, `idxD`).

5 Reproduction

`python3 reproduce.py` (standard library only, no third-party packages) rebuilds $H_n(0)$ for $n = 2, \dots, 5$, re-runs the three self-checks, recomputes Table 1, prints the six principal two-sided ideals of $H_3(0)$ and the hook-length bookkeeping for the partitions of 3. Its output agrees cell-by-cell with the Lean computation.

References

- [1] O. Ganyushkin and V. Mazorchuk, *On Kiselman quotients of 0-Hecke monoids*, Int. Electron. J. Algebra **10** (2011), 174–204; arXiv:1006.0316. (The $\mathcal{J}$ -triviality of H_Γ is derived there.)